WARFARIN DOSE MODELING STUDY

IWPC calculator benchmarking with machine-learning baselines and explainability

Shared 20% holdout evaluation on the IWPC cohort. Models compared: IWPC clinical formula, IWPC pharmacogenetic calculator, linear regression, random forest, XGBoost, neural network, and the tuned LightGBM repository model.

BEST MODEL

Our Model (Tuned LightGBM)

RMSE improved by **0.02** mg/week over the IWPC pharmacogenetic calculator, with a modest **0.3** point gain in within-20% accuracy.

Best RMSE: 11.90 mg/week

Best MAE: 8.70 mg/week

Best R^2: 0.421

Clinical-only IWPC: RMSE 13.84, MAE 10.07

SOURCE COHORT: IWPC SUBJECT DATA

SLIDE 01

LEADERBOARD

Top three models are effectively clustered, but the repository model lands first on the primary metric

01

Our Model (Tuned LightGBM)

RMSE	MAE
11.90	8.70

R^2	WITHIN 20%
0.421	42.0%

02

IWPC Pharmacogenetic Calculator

RMSE	MAE
11.92	8.65

R^2	WITHIN 20%
0.419	41.7%

03

Linear Regression

RMSE	MAE
11.92	8.74

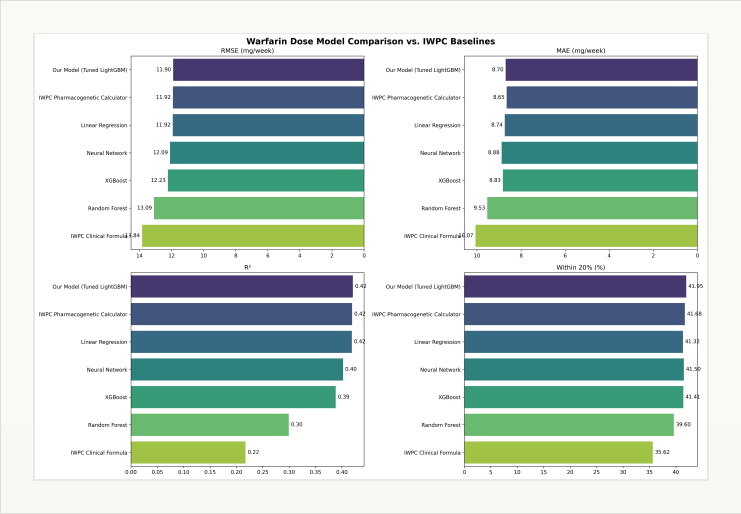
R^2	WITHIN 20%
0.419	41.3%

METRIC PRIORITY: RMSE, THEN MAE

SLIDE 02

VISUAL COMPARISON

Aggregate metrics and patient-level fit show a narrow edge over the IWPC pharmacogenetic calculator



The tuned LightGBM model is first by RMSE and R^2, but the IWPC pharmacogenetic formula remains highly competitive.

Linear regression nearly matches the top two, suggesting the signal is still dominated by strong structured clinical-genetic effects.

Random forest underperforms on this cohort and should not be positioned as a preferred baseline here.

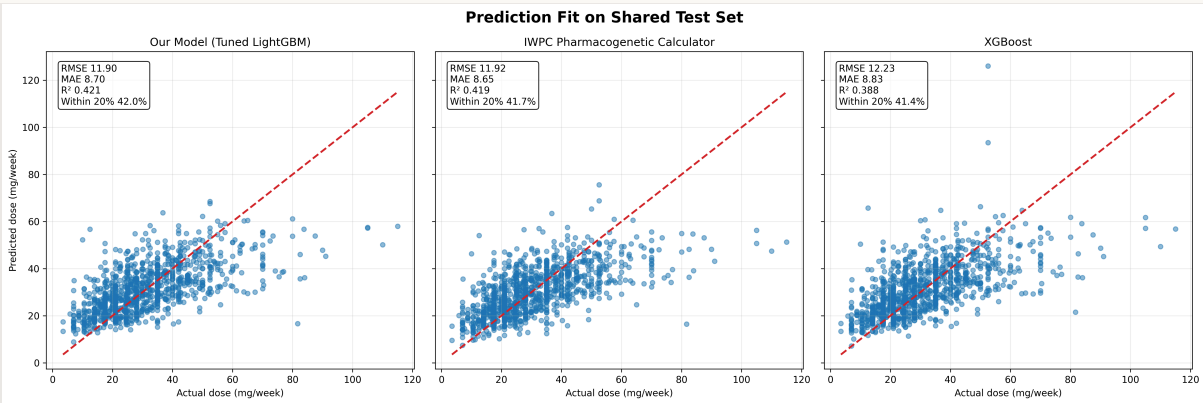
The clinical-only IWPC formula shows a clear step down, which justifies retaining genotype-aware modeling in the pipeline.

IMAGES GENERATED FROM REPOSITORY OUTPUTS

SLIDE 03

FIT DIAGNOSTICS

Patient-level scatter plots show similar calibration bands for the top genotype-aware approaches

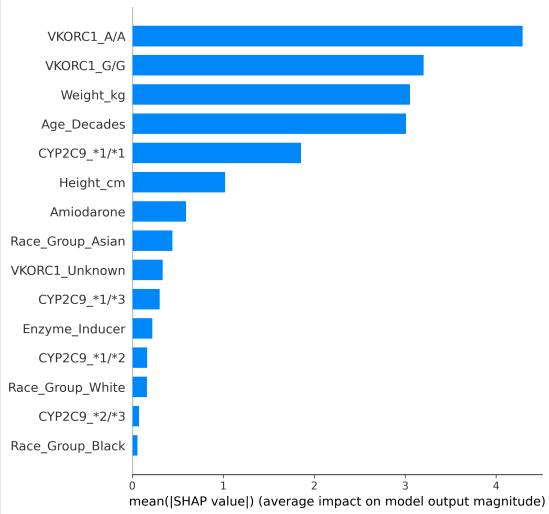


SHARED HOLDOUT SPLIT: RANDOM_STATE 42

SLIDE 04

EXPLAINABILITY

SHAP confirms the learned model still tracks clinically credible genetic and anthropometric drivers



Global feature influence is led by **VKORC1**, then **weight**, **age**, and **CYP2C9**.

The strongest SHAP features align with known warfarin dose biology rather than opaque proxy variables.

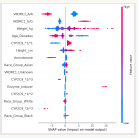
This makes the best model easier to defend clinically than a purely black-box ranking would suggest.

VKORC1 A/A	**4.292**
VKORC1 G/G	**3.202**
Weight kg	**3.053**
Age Decades	**3.011**
CYP2C9 *1/*1	**1.855**
Height cm	**1.021**
Amiodarone	**0.589**
Race Group Asian	**0.440**

APPENDIX

Presentation-ready result table and SHAP distribution view

RANK	MODEL	RMSE	MAE	R^2	WITHIN 20%
1	Our Model (Tuned LightGBM)	11.9034	8.7012	0.4209	41.9530
2	IWPC Pharmacogenetic Calculator	11.9198	8.6526	0.4193	41.6817
3	Linear Regression	11.9238	8.7364	0.4190	41.3201
4	Neural Network	12.0937	8.8810	0.4023	41.5009
5	XGBoost	12.2332	8.8270	0.3884	41.4105
6	Random Forest	13.0928	9.5326	0.2994	39.6022
7	IWPC Clinical Formula	13.8394	10.0696	0.2173	35.6239



GENERATED FROM IWPC_MODEL_METRICS.CSV AND IWPC_SHAP_TOP_FEATURES.CSV

SLIDE 06